

HDOC Day-14

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SUBJECT: C programming

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Question 1: Check Whether All Digits Are Unique

Write a C program to input a positive integer and check whether all its digits are unique.

Algorithm

1. Start.
2. Input a positive integer n.
3. Initialize freq[10] to 0 and set unique = 1.
4. Extract each digit using $n \% 10$ and increase its frequency.
5. If any digit occurs more than once, set unique = 0.
6. Display "Unique" if unique = 1; otherwise display "Not Unique".
7. Stop.

Test Case Table

Input	Expected Output
12345	Unique
12234	Not Unique
9876543210	Unique
1002	Not Unique

C Program

```
#include <stdio.h>

int main()
{
    int n, temp, digit;
    int freq[10] = {0};
    int unique = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &n);
    temp = n;

    while (temp > 0)
    {
        digit = temp % 10;
        freq[digit]++;
        if (freq[digit] > 1)
        {
            unique = 0;
            break;
        }
        temp = temp / 10;
    }

    if (unique == 1)
        printf("Unique\n");
    else
```

```

printf("Not Unique\n");

return 0;
}

```

Compile

```
gcc uniquedigit.c -lm
```

Execute

```
./a.out
```

Testing

Input	Result
12345	Unique
12234	Not Unique
987654321	Unique
1002	Not Unique

```

(kali㉿kali)-[~/Desktop/cprog]
$ gcc uniquedigit.c -lm

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer:12345
Unique

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer:12234
Not Unique

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer:987654321
Unique

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer:1002
Not Unique

```

Question 2: Find Frequency of Each Digit from 0 to 9

Write a C program to input a positive integer and find the frequency of each digit from 0 to 9.

Algorithm

- Start.
- Input a positive integer n.
- Initialize freq[10] to 0.
- Extract each digit using $n \% 10$.
- Increase freq[digit] and remove the last digit using $n / 10$.
- Repeat until all digits are processed.
- Display the frequency of digits 0 to 9.
- Stop.

Test Case Table

Input	Expected Frequency
112333	0:0, 1:2, 2:1, 3:3, others:0

12345	1-5:1, others:0
9999	9:4, others:0
1002	0:2, 1:1, 2:1, others:0

C Program

```
#include <stdio.h>

int main()
{
    int n, temp, digit, i;
    int freq[10] = {0};

    printf("Enter a positive integer: ");
    scanf("%d", &n);
    temp = n;

    while (temp > 0)
    {
        digit = temp % 10;
        freq[digit]++;
        temp = temp / 10;
    }

    for (i = 0; i <= 9; i++)
        printf("freq %d = %d\n", i, freq[i]);

    return 0;
}
```

Compile

```
gcc frequency.c -lm
```

Execute

```
./a.out
```

Testing

Input	Result
112333	0:0, 1:2, 2:1, 3:3, others:0
12345	1-5:1, others:0
9999	9:4, others:0
1002	0:2, 1:1, 2:1, others:0

```

(kali@kali)-[~/Desktop/cprog]
$ gcc frequency.c -lm
(kali@kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 112333
freq 0 = 0
freq 1 = 2
freq 2 = 1
freq 3 = 3
freq 4 = 0
freq 5 = 0
freq 6 = 0
freq 7 = 0
freq 8 = 0
freq 9 = 0
(kali@kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 12345
freq 0 = 0
freq 1 = 1
freq 2 = 1
freq 3 = 1
freq 4 = 1
freq 5 = 1
freq 6 = 0
freq 7 = 0
freq 8 = 0
freq 9 = 0

```

```

(kali@kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 9999
freq 0 = 0
freq 1 = 0
freq 2 = 0
freq 3 = 0
freq 4 = 0
freq 5 = 0
freq 6 = 0
freq 7 = 0
freq 8 = 0
freq 9 = 4
(kali@kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 1002
freq 0 = 2
freq 1 = 1
freq 2 = 1
freq 3 = 0
freq 4 = 0
freq 5 = 0
freq 6 = 0
freq 7 = 0
freq 8 = 0
freq 9 = 0

```